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Seminar 1
ANALIZĂ MATEMATICĂ

TITLU: RECAPITULARE ȘIRURI

① **Monotonie:** $\frac{a_{n+1}}{a_n} = \frac{1\cdot3\cdot5\cdot\ldots\cdot(2n+1)}{2\cdot4\cdot6\cdot\ldots\cdot(2n+2)} \cdot \frac{2\cdot4\cdot6\cdot\ldots\cdot(2n)}{1\cdot3\cdot5\cdot\ldots\cdot(2n-1)} = \frac{2n+1}{2n+2} < 1 \left(\forall n \in \mathbb{N}^*\right)$

$\Rightarrow$ șirul $(a_n)_{n\in\mathbb{N}}$ este strict descrescător

~~Cum șirul $(a_n)_{n\in\mathbb{N}}$ este strict descrescător $\Rightarrow a_n < a_1$ ...~~

Mărginire: Cum șirul $(a_n)_{n\in\mathbb{N}}$ este strict descrescător și fiecare termen este strict pozitiv, deci are margine inferioară (intuitiv și margine superioară, deoarece orice șir strict descrescător are margine superioară)

TW $\Rightarrow (a_n)_{n\in\mathbb{N}}$ este convergent și are limită la infinit.

~~Exam~~

$a_n = \frac{1}{2}\cdot\frac{3}{4}\cdot\frac{5}{6}\cdot\ldots\cdot\frac{2n-1}{2n}$

$\frac{1}{2}\cdot\frac{2}{3}\cdot\frac{3}{4}\cdot\ldots\cdot\frac{2n-1}{2n} = \frac{1}{2}\cdot\frac{2}{3}\cdot\frac{4}{5}\cdot\ldots\cdot\frac{2n-2}{2n-1} < a_n < \frac{2}{3}\cdot\frac{4}{5}\cdot\frac{6}{7}\cdot\ldots\cdot\frac{2n}{2n+1} \quad \Big| \cdot a_n$

$\left(\frac{1}{2}\cdot\frac{2}{3}\cdot\frac{4}{5}\cdot\ldots\cdot\frac{2n-2}{2n-1}\right)\left(\frac{1}{2}\cdot\frac{3}{4}\cdot\frac{5}{6}\cdot\ldots\cdot\frac{2n-1}{2n}\right) < a_n^2 < \left(\frac{1}{2}\cdot\frac{2}{3}\cdot\frac{4}{5}\cdot\ldots\cdot\frac{2n-1}{2n}\right)\left(\frac{2}{3}\cdot\frac{4}{5}\cdot\frac{6}{7}\cdot\ldots\cdot\frac{2n}{2n+1}\right)$

$\Rightarrow \frac{1}{4n} < a_n^2 < \frac{1}{2n+1} \overset{a_n>0}{\Rightarrow} \frac{1}{2\sqrt{n}} < a_n < \frac{1}{\sqrt{2n+1}}$ (criteriul cleștelui)

$n \to +\infty$

$\Rightarrow \lim\limits_{n\to+\infty} a_n = 0$

② $a_n = \frac{n^n}{(n!)^2}$

Monotonie: $\frac{a_{n+1}}{a_n} = \frac{(n+1)^{(n+1)}}{[(n+1)!]^2} \cdot \frac{(n!)^2}{n^n} = \left(\frac{n+1}{n}\right)^n \cdot \frac{1}{n+1} =$

$= \frac{1}{n+1}\left(1+\frac{1}{n}\right)^n < \frac{e}{n+1} < 1 \left(\forall n \in \mathbb{N}^* - \{1\}\right)$ ①

$\left(1+\frac{1}{n}\right)^n \to e$

Pentru $n=1$, $\frac{a_{n+1}}{a_n} = \frac{1}{2}\cdot 2 = 1 \Rightarrow a_1 = a_2$ ②

Din ①, ② $\Rightarrow (a_n)_{n\in\mathbb{N}}$ este descrescător I

Cum $(a_n)_{n\in\mathbb{N}}$ are elem. strict pozitive $\Rightarrow a_n > 0 \left(\forall n \in \mathbb{N}^*\right) \Rightarrow$ $(a_n)_{n\in\mathbb{N}}$ este mărg. de inf. și cum $(a_n)_{n\in\mathbb{N}}$ este descrescător $\Rightarrow$ $(a_n)_{n\in\mathbb{N}}$ este mărg. sup. $\Rightarrow (a_n)_{n\in\mathbb{N}}$ este mărg. II

2/ Dim I, II $\overset{T.W.}{\Rightarrow}$ $(a_n)_{n\in\mathbb{N}}$ este convergent

$\lim\limits_{n\to+\infty} \frac{a_{n+1}}{a_n} = \lim\limits_{n\to+\infty} \underbrace{\frac{1}{n+1}}_{\downarrow 0} \underbrace{\left(1+\frac{1}{n}\right)^n}_{\downarrow e} = 0 < 1 \Rightarrow$ conform criteriului raportului $\Rightarrow$

$$\boxed{\lim\limits_{n\to+\infty} a_n = 0}$$

3. $\lim\limits_{n\to+\infty} \frac{n^k}{a^n} = x_n$

$$\frac{x_{n+1}}{x_n} = \frac{(n+1)^k}{a^{n+1}} \cdot \frac{a^n}{n^k} = \left(\frac{n+1}{n}\right)^k \cdot \frac{1}{a}$$

$$\Rightarrow \lim_{n\to+\infty} \frac{x_{n+1}}{x_n} = \lim_{n\to+\infty} \underbrace{\left(\frac{n+1}{n}\right)^k}_{\downarrow 1} \cdot \frac{1}{a} = \frac{1}{a}$$

Dacă $\frac{1}{a} > 1 \Leftrightarrow a < 1 \Rightarrow$ conform crit. raportului $\Rightarrow \lim\limits_{n\to+\infty} x_n = +\infty$

Dacă $\frac{1}{a} < 1 \Leftrightarrow a > 1 \Rightarrow$ —u— $\Rightarrow \lim\limits_{n\to+\infty} x_n = 0$

Dacă $a = 1 \Rightarrow \lim\limits_{n\to+\infty} n^k = +\infty$

(4) $\lim\limits_{n\to+\infty} \frac{\alpha^n}{n!}$

$$a_n = \frac{\alpha^n}{n!}$$

$$\lim_{n\to+\infty} \frac{a_{n+1}}{a_n} = \lim_{n\to+\infty} \frac{\alpha^{n+1}}{(n+1)!} \cdot \frac{n!}{\alpha^n} = \lim_{n\to+\infty} \frac{\alpha}{n+1} = 0 \overset{\text{conform}}{\Rightarrow} \text{crit. raportului}$$

$$\lim_{n\to+\infty} \frac{\alpha^n}{n!} = 0$$

(5) $\lim\limits_{n\to+\infty} \sqrt[3]{n}\cdot\left(\sqrt[3]{(n+1)^2} - \sqrt[3]{(n-1)^2}\right) = \lim\limits_{n\to+\infty} \sqrt[3]{n} \cdot \frac{(n+1)^2 - (n-1)^2}{(n+1)^{\frac{4}{3}} + [(n+1)(n-1)]^{\frac{2}{3}} + (n-1)^{4/3}}$

$= \lim\limits_{n\to+\infty} \frac{\sqrt[3]{n}\,[(n+1)^2 - \sqrt[3]{n}\cdot 4n}{(n+1)^{\frac{4}{3}} + [(n+1)(n-1)]^{\frac{2}{3}} + (n-1)^{\frac{4}{3}}} = \frac{4}{3}$

(6.) $\lim\limits_{n\to+\infty} n^k\cdot\left(\sqrt{\frac{n+2}{n+5}} - 1\right) = \lim\limits_{n\to+\infty} \left(n^k \cdot \frac{\frac{n+2}{n+5} - 1}{\sqrt{\frac{n+2}{n+5}} + 1}\right) = \lim\limits_{n\to+\infty} \left(n^k \cdot \frac{n+2-(n+5)}{(n+5)\left(\sqrt{\frac{n+2}{n+5}} + 1\right)}\right) =$

$= \lim\limits_{n\to+\infty} n^k \cdot \frac{-3}{(n+5)\left(\sqrt{\frac{n+2}{n+5}} + 1\right)} = L$

Dacă $k = 1 \Rightarrow L = \lim\limits_{n\to+\infty} \frac{-3n}{(n+5)\left(\sqrt{\frac{n+2}{n+5}} + 1\right)} = \frac{-3}{2}$

Dacă $k > 1 \Rightarrow L = -\infty$

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(7.) $\sum_{k=1}^{n} \frac{\sin k}{n^2+k} = a_n$, $n \in \mathbb{N}^*$

$\Rightarrow \quad -1 \le \sin k \le 1, \ k \in \overline{1,n} \Rightarrow$

$\sum_{k=1}^{n} \frac{-1}{n^2+k} < a_n < \sum_{k=1}^{n} \frac{1}{n^2+k} \Rightarrow 0 < |a_n| < \sum_{k=1}^{n} \frac{1}{n^2+k} < \sum_{k=1}^{n} \frac{1}{n^2+1} = \frac{n}{n^2+1} \Rightarrow$

$\frac{n}{n^2+1} \to 0$

$\lim_{n\to+\infty}$ conform criteriului majorării : $\lim_{n\to+\infty} a_n = 0$

(8.) $\lim_{n\to+\infty} \frac{1+2\sqrt{2}+3\sqrt{3}+\dots+n\sqrt{n}}{n^2\sqrt{n}}$

Fie $(a_n)_{n\in\mathbb{N}}, (b_n)_{n\in\mathbb{N}}$, $a_n = \sum_{k=1}^{n} k\sqrt{k}$, $b_n = n^2\sqrt{n}$

Din teorema lui Stolz avem :

$\lim_{n\to+\infty} \frac{a_n}{b_n} = \lim_{n\to+\infty} \frac{a_{n+1}-a_n}{b_{n+1}-b_n} = \lim_{n\to+\infty} \frac{(n+1)\sqrt{n+1}}{(n+1)^2\sqrt{n+1} - n^2\sqrt{n}} =$

$= \lim_{n\to+\infty} \frac{(n+1)\sqrt{n+1}\left[(n+1)^2\sqrt{n+1} + n^2\sqrt{n}\right]}{(n+1)^5 - n^5} = \lim_{n\to+\infty} \frac{(n+1)\sqrt{n+1}\left[(n+1)^2\sqrt{n+1} + n^2\sqrt{n}\right]}{C_1^5 n^4 + \dots}$

$= \frac{2}{5}$

(9) ~~$\lim_{n\to+\infty} (\sqrt[3]{n^3+n+x} \cdot \pi) =$~~

~~$\lim_{n\to+\infty} \sin(\sqrt[3]{n^3+n+x} \cdot \pi) = \lim_{n\to+\infty} \sin(\sqrt[3]{n^3+n+x} \cdot \pi - \pi + \pi) =$~~

~~$= \lim_{n\to+\infty} (-1) \cdot \sin(\pi(\sqrt[3]{n^3+n+x} - 1)) = \lim_{n\to+\infty} (-1) \cdot \sin\left(\pi\left(\frac{n^3+n+x-1}{\dots}\right)\right)$~~

(10.) $\lim_{n\to+\infty} \sin(n\pi\sqrt[n]{e}) = \lim_{n\to+\infty} \sin(n\pi\sqrt[n]{e} - n\pi + n\pi) =$

$\lim_{n\to+\infty} \sin(n\pi\sqrt[n]{e} - n\pi)\cdot(-1)^n = \lim_{n\to+\infty} (-1)^n \cdot \lim\left(n\pi \cdot \frac{e^{\frac{1}{n}} - 1}{1/n}\right) = 0$

mărg. $\to 0$, $\frac{1}{n} \to 0$, $\to 1$ $\to$ limită fundamentală

(9) $\lim_{n\to+\infty} \sin(\sqrt[3]{n^3+n+x} \cdot \pi) = \lim_{n\to+\infty} \sin(\sqrt[3]{n^3+n+x} \cdot \pi - n\pi + n\pi) =$

$= \lim_{n\to+\infty} (-1)^n \cdot \sin(\sqrt[3]{n^3+n+x} \cdot \pi - n\pi) = \lim_{n\to+\infty} (-1)^n \cdot \sin\left(\pi \cdot \left[\sqrt[3]{n^3+n+x} - n\right]\right) =$

~~$= \lim_{n\to+\infty} (-1)^n \cdot \sin\left[\pi \cdot \frac{n^3+n+x-n^3}{\sqrt[3]{n^3+n+x}+n}\right] = \lim_{n\to+\infty} (-1)^n \cdot \sin\left(\pi \cdot \frac{n+x}{\sqrt[3]{n^3+n+x}+n}\right)$~~

$= \lim_{n\to+\infty} (-1)^n \cdot \sin\left(\pi \cdot \frac{n^3+n+x-n^3}{(n^3+n+x)^{\frac{2}{3}} + n\sqrt[3]{n^3+n+x} + n^2}\right) = \lim_{n\to+\infty} (-1)^n \cdot \sin\left(\pi \cdot \frac{n+x}{(n^3+n+x)^{\frac{2}{3}} + n\sqrt[3]{n^3+n+x} + n^2}\right)$

$\to 0$

$= 0$

(11.) Întrucât $\boxed{\left(1+\frac{1}{n}\right)^n < e < \left(1+\frac{1}{n}\right)^{n+1} \quad (\forall) n \in \mathbb{N}^*}$ $\Rightarrow$

$$\Rightarrow \sqrt{n}\cdot\left[\left(1+\frac{1}{n}\right)^n-\left(1+\frac{1}{n}\right)^n\right] < \sqrt{n}\cdot\left[e-\left(1+\frac{1}{n}\right)^n\right] < \sqrt{n}\cdot\left[\left(1+\frac{1}{n}\right)^{n+1}-\left(1+\frac{1}{n}\right)^n\right]$$

$$\Rightarrow \quad 0 < \sqrt{n}\cdot\left[e-\left(1+\frac{1}{n}\right)^n\right] < \sqrt{n}\cdot\left(1+\frac{1}{n}-1\right)\cdot\left(1+\frac{1}{n}\right)^n$$

$$\Rightarrow \quad 0 < \sqrt{n}\cdot\left[e-\left(1+\frac{1}{n}\right)^n\right] < \sqrt{n}\cdot\frac{1}{n}\cdot\left(1+\frac{1}{n}\right)^n$$

($\sqrt{n}\cdot\frac{1}{n} \to 0$, $\left(1+\frac{1}{n}\right)^n \to e$)

$\Rightarrow$ conform criteriului cleștelui $\Rightarrow \lim\limits_{n\to+\infty}\left(\sqrt{n}\cdot\left[e-\left(1+\frac{1}{n}\right)^n\right]\right)=0$

(12.) $\lim\limits_{n\to+\infty}\left(\frac{1}{2n+1}+\frac{1}{2n+2}+\ldots+\frac{1}{3n}\right) = \lim\limits_{n\to+\infty}\left(1+\frac{1}{2}+\ldots+\frac{1}{3n}\right) -$

$\lim\limits_{n\to+\infty}\left(1+\frac{1}{2}+\ldots+\frac{1}{2n}\right) = \lim\limits_{n\to+\infty}\left(1+\frac{1}{2}+\ldots+\frac{1}{3n}-\ln 3n\right) - \lim\limits_{n\to+\infty}\left(1+\frac{1}{2}+\ldots+\frac{1}{2n}-\ln 2n\right)$

$+ \lim\limits_{n\to+\infty}(\ln 3n - \ln 2n) = c - c + \lim\limits_{n\to+\infty}\ln\frac{3n}{2n} = \ln\frac{3}{2}$

(13.) Notăm $a_n = \frac{(n!)^2}{(2n)!}$, $n\in\mathbb{N}^*$

$$\lim_{n\to+\infty}\frac{a_{n+1}}{a_n} = \lim_{n\to+\infty}\frac{((n+1)!)^2}{(2n+2)!}\cdot\frac{(2n)!}{(n!)^2} = \lim_{n\to+\infty}\frac{(n+1)^2}{(2n+2)(2n+1)} = \frac{1}{4}$$

Cum $\lim\limits_{n\to+\infty}\frac{a_{n+1}}{a_n} = \frac{1}{4} \Rightarrow \lim\limits_{n\to+\infty}\sqrt[n]{a_n} = \lim\limits_{n\to+\infty}\sqrt[n]{\frac{(n!)^2}{(2n)!}} = \frac{1}{4}$

(14.) Notăm $a_n = \frac{3^{3n}(n!)^3}{(3n)!}$, $n\in\mathbb{N}^*$

$$\lim_{n\to+\infty}\frac{a_{n+1}}{a_n} = \lim_{n\to+\infty}\frac{3^{3n+3}[(n+1)!]^3}{(3n+3)!}\cdot\frac{(3n)!}{3^{3n}(n!)^3} =$$

$$= \lim_{n\to+\infty}\frac{3^3\cdot(n+1)^3}{(3n+1)(3n+2)(3n+3)} = 1.$$

$\Rightarrow$ Cum $\lim\limits_{n\to+\infty}\frac{a_{n+1}}{a_n} = 1 \Rightarrow \lim\limits_{n\to+\infty}\sqrt[n]{a_n} = \lim\limits_{n\to+\infty}\sqrt[n]{\frac{3^{3n}(n!)^3}{(3n)!}} = 1$

(15.) $\lim\limits_{n\to+\infty}\sum\limits_{k=1}^{n}\frac{k^2+k+1}{(k+1)!} = \lim\limits_{n\to+\infty}\sum\limits_{k=1}^{n}\frac{k(k+1)-1}{(k+1)!} = \frac{1}{2}+\lim\limits_{n\to+\infty}\sum\limits_{k=2}^{n}\left(\frac{1}{(k-1)!}-\frac{1}{(k+1)!}\right) =$

$$= \frac{1}{2}+\lim_{n\to+\infty}\left(1-\cancel{\frac{1}{3!}}+\frac{1}{2!}-\cancel{\frac{1}{4!}}+\cancel{\frac{1}{3!}}-\cancel{\frac{1}{5!}}+\ldots+\cancel{\frac{1}{(n-1)!}}-\frac{1}{(n+1)!}\right) = \frac{1}{2}+\frac{1}{2}+1-\lim_{n\to+\infty}\left(\underbrace{\frac{1}{n!}}_{\to 0}+\underbrace{\frac{1}{(n+1)!}}_{\to 0}\right)$$

$$= \frac{3}{2}+\frac{1}{2} = 2$$

(16.) $\lim_{n\to+\infty} \sum_{k=1}^{n} \frac{2^{k-1}(k-1)}{(k+1)!} = \lim_{n\to+\infty} \sum_{k=1}^{n} \frac{2^{k-1}(k+1-2)}{(k+1)!} =$

$\lim_{n\to+\infty} \sum_{k=1}^{n} \left(\frac{2^{k-1}\cdot(k+1)}{(k+1)!} - \frac{2^{k-1}\cdot 2}{(k+1)!} \right) = \lim_{n\to+\infty} \sum_{k=1}^{n} \left(\frac{2^{k-1}}{k!} - \frac{2^k}{(k+1)!} \right)$

$= \lim_{n\to+\infty} \left(\frac{2^0}{1!} - \cancel{\frac{2^1}{2!}} + \cancel{\frac{2^1}{2!}} + \cancel{\frac{2^2}{3!}} + \ldots + \cancel{\frac{2^{n-1}}{n!}} - \frac{2^n}{(n+1)!} \right)$

$= \lim_{n\to+\infty} \left(1 + \left(-\frac{2^n}{(n+1)!} \right) \right) = 1 + \lim_{n\to+\infty} \frac{\overbrace{2\cdot 2\cdot \ldots \cdot 2}^{n}}{1\cdot 2\cdot \ldots \cdot n} = 1$

$\downarrow 0$

(17.) Notăm $a_n = \sum_{k=1}^{n} \left(\sqrt{1+\frac{k}{n^2}} - 1 \right)$

$a_n = \sum_{k=1}^{n} \left(\frac{1+\frac{k}{n^2}-1}{\sqrt{1+\frac{k}{n^2}}+1} \right) = \sum_{k=1}^{n} \frac{\frac{k}{n^2}}{\sqrt{1+\frac{k}{n^2}}+1} \Rightarrow$

$\sum_{k=1}^{n} \frac{1}{n^2}\cdot\frac{k}{\sqrt{1+\frac{1}{n}}+1} < a_n < \sum_{k=1}^{n} \frac{1}{n^2}\cdot\frac{k}{\sqrt{1+\frac{1}{n^2}}+1} \Rightarrow$

$\sum_{k=1}^{n}$ $\frac{n(n+1)}{2}\cdot\frac{1}{n^2}\cdot\frac{1}{\sqrt{1+\frac{1}{n}}+1} < a_n < \frac{n(n+1)}{2}\cdot\frac{1}{n^2}\cdot\frac{1}{\sqrt{1+\frac{1}{n^2}}+1}$

$\to 1/2$ $\to 1/2$ $\to 1/2$ $\to 1/2$

$\frac{1}{n} \to 0$ $\quad \frac{1}{n^2} \to 0$

$\Rightarrow$ conform criteriului clestelui :

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$\lim_{n\to+\infty} a_n = 1/4$

(18.) $\lim_{n\to+\infty} \left(1-\frac{1}{3}\right)\left(1-\frac{1}{6}\right)\cdot\ldots\cdot\left(1-\frac{1}{C_{n+1}^2}\right) =$

$= \lim_{n\to+\infty} \prod_{k=2}^{n} \left(1-\frac{1}{C_{n+1}^2}\right) = \lim_{n\to+\infty} \prod_{k=2}^{n} \left(1-\frac{1}{\frac{(k+1)!}{(k-1)!\,2}}\right) =$

$= \lim_{n\to+\infty} \prod_{k=2}^{n} \left(1-\frac{2}{k(k+1)}\right) = \lim_{n\to+\infty} \prod_{k=2}^{n} \frac{k(k+1)-2}{k(k+1)} =$

$= \lim_{n\to+\infty} \prod_{k=2}^{n} \frac{k^2+k-2}{k(k+1)} = \lim_{n\to+\infty} \prod_{k=2}^{n} \frac{(k-1)(k+2)}{k(k+1)} =$

$$= \lim_{n\to+\infty} \frac{1\cdot 1}{1\cdot 2}\cdot\frac{2\cdot 5}{2\cdot 3} = \lim_{n\to+\infty} \prod_{k=2}^{n} \frac{k-1}{k}\cdot\frac{k+2}{k+1} =$$

$$= \lim_{n\to+\infty}\left(\frac{1}{2}\cdot\frac{4}{3}\cdot\frac{2}{3}\cdot\frac{5}{4}\cdot\frac{3}{4}\cdot\frac{6}{5}\cdot\ldots\cdot\frac{n-1}{n}\cdot\frac{n+2}{n+1}\right) =$$

$$= \lim_{n\to+\infty}\frac{n+2}{n}\cdot\frac{1}{3} = 1/3$$

(19) $$\lim_{n\to+\infty}\frac{C_n^1\cdot C_n^2\cdot\ldots\cdot C_n^p}{p^{\frac{p(p+1)}{2}}} = \lim_{n\to+\infty}\frac{\frac{n!}{(n-1)!}\cdot\frac{n!}{(n-2)!2!}\cdot\ldots\cdot\frac{n!}{(n-p)!p!}}{n^{1+2+\ldots+p}}$$

$$= \lim_{n\to+\infty}\frac{n\cdot\frac{n(n-1)}{2!}\cdot\frac{n(n-1)(n-2)}{3!}\cdot\ldots\cdot\frac{n(n-1)\cdot\ldots\cdot(n-p+1)}{p!}}{n\cdot n^2\cdot n^3\cdot\ldots\cdot n^p} =$$

$$= \lim_{n\to+\infty}\frac{n}{n}\cdot\frac{n(n-1)}{n^2\cdot 2!}\cdot\frac{n(n-1)(n-2)}{n^3\cdot 3!}\cdot\ldots\cdot\frac{n(n-1)\cdot\ldots\cdot(n-p+1)}{n^p\cdot p!}$$

$$= \frac{1}{2!\cdot 3!\cdot\ldots\cdot p!}$$

(20) $$\lim_{n\to+\infty}\frac{2^2\cdot 4^2\cdot\ldots\cdot(2n)^2}{1^2\cdot 3^2\cdot 5^2\cdot\ldots\cdot(2n-1)^2}\cdot\frac{1}{\sqrt[n]{n!}} \overset{\text{F. Willis}}{=\!=} \lim_{n\to+\infty}\frac{\pi}{2}\cdot\frac{2n+1}{\sqrt[n]{n!}} =$$

$$= \lim_{n\to+\infty}\sqrt[n]{\frac{(2n+1)^n}{n!}}\cdot\frac{\pi}{2} \quad (1)$$

Notăm $a_n = \frac{(2n+1)^n}{n!}$

$$\lim_{n\to+\infty}\frac{a_{n+1}}{a_n} = \lim_{n\to+\infty}\frac{(2n+3)^{n+1}}{(n+1)!}\cdot\frac{n!}{(2n+1)^n} = \lim_{n\to+\infty}\frac{(2n+3)^{n+1}}{(2n+1)^n}\cdot\frac{1}{n+1}$$

$$= \lim_{n\to+\infty}\left(\frac{2n+3}{2n+1}\right)^n\cdot\frac{2n+3}{n+1} = \lim_{n\to+\infty}\left[\left(1+\frac{2}{2n+1}\right)^{\frac{2n+1}{2}}\right]^{\frac{2n}{2n+1}} :2 =$$

$\frac{2n+3}{n+1} \to 2$; $\frac{2n}{2n+1} \to 1$; $\left(1+\frac{2}{2n+1}\right) \to e$

$= 2e \Rightarrow$

Cum $\lim_{n\to+\infty}\frac{a_{n+1}}{a_n} = 2e \Rightarrow \lim_{n\to+\infty}\sqrt[n]{a_n} = 2e \Rightarrow$ (1)

$$\lim_{n\to+\infty}\frac{(2n)!!}{(2n-1)!!}\cdot\frac{1}{\sqrt[n]{n!}} = 2e\cdot\frac{\pi}{2} = e\pi$$